

MEDI_603

MEDICAL PHARMACOLOGY & TOXICOLOGY



COURSE SYLLABUS 2024 - 2025

Medical Pharmacology and Toxicology

MEDI_603

2024-2025 Syllabus

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Required Textbook and Alternative Sources of Information

The required course textbook is: **Katzung's Basic & Clinical Pharmacology**, by Todd W. Vanderah, 16TH Ed., 2024, McGraw Hill, NY. (Available from Access Medicine at [Katzung's Basic & Clinical Pharmacology, 16th Edition | AccessMedicine | McGraw Hill Medical \(mhmedical.com\)](#)). You are required to read this primary source of information prior to attending lecture or attempting to answer practice test questions. In addition, there is a wide variety of current, well-written and well-illustrated general textbooks in pharmacology. You are encouraged to purchase other pharmacology books of your choice to use for reference and study. Some texts have test question sets or companion texts with review notes and test questions such as **Katzung's Pharmacology: Examination & Board Review, Fourteenth Edition** (Available from Access Medicine at [Katzung's Pharmacology Examination & Board Review, 14th Edition | AccessMedicine | McGraw Hill Medical \(mhmedical.com\)](#)).

Other Recommended Pharmacology Textbooks

Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy Fourth Edition, 4e David E. Golan, Ehrin J. Armstrong, April W. Armstrong ([Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy Fourth Edition, 4e | Premium Basic Sciences | Health Library \(lwwhealthlibrary.com\)](#))

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, Laurence L. Brunton, Björn C. Knollmann, 2023 McGraw-Hill, New York, NY; ISBN 978-1-264-25807-9. ([Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e | AccessMedicine | McGraw Hill Medical \(mhmedical.com\)](#))

Rang & Dale's Pharmacology, 10th Edition (2023), James M. Ritter, Rod J. Flower, et al., ISBN: 9780323873956; Elsevier

Brody's Human Pharmacology: Mechanism-Based Therapeutics 6th Edition, Elsevier.

Review Books:

Several abbreviated review books in pharmacology can be used. One particularly favored by students is: **Lippincott's Illustrated Reviews - Pharmacology** ([Lippincott® Illustrated Reviews: Pharmacology, 8e | Premium Basic Sciences | Health Library \(lwwhealthlibrary.com\)](#)) - updated periodically, published by Lippincott-Raven. Other review books include BRS Pharmacology ([BRS Pharmacology, 7e | Board Review Series | Health Library \(lwwhealthlibrary.com\)](#)) and "First Aid" published by McGraw Hill.

There are a number of USMLE review books and computer-based test products.

The Department of Pharmacology and Toxicology encourages the use of these review materials for review and supplementary study but **cautions that review books do not provide the quality of in-depth information found in primary, more comprehensive textbooks**. Students are encouraged to utilize comprehensive subject area test question books as a source for personal assessment of your knowledge of pharmacology throughout the course and for USMLE preparation.

Other Text Resources:

In light of the curriculum being discipline-integrated, there are a number of useful cross-discipline reference sources. An example of a book that covers broad-based information includes **Current Medical Diagnosis and Treatment 2025**, McGraw Hill, NY, available on MSM's library intranet (<https://accessmedicine.mhmedical.com/book.aspx?bookid=3495>). Texts of this nature make excellent references for pharmacology, pathology, and pathophysiology. As interest in self-

medicating with natural products continues to grow, physicians must keep up with alternative medicinal products. **Tyler's Herbs of Choice: The Therapeutic Use of Phytomedicinals**, D.V.C. Awang, CRC Press, NY, 3rd ed., 2009 is a current work in this subject matter.

Course Material/Handouts, etc.

Faculty will post lecture slides on the Medical Pharmacology Canvas intranet site. In many cases, individual faculty will provide students with review exam question sets, exercises and other instructional material. Please check the Medical Pharmacology Canvas site frequently for messages and course-related information. The use of the MSM intranet service for purposes of instruction and communication is at the discretion of individual faculty.

Internet-based instructional material:

The faculty recognizes there is limited time that a student can devote to studying an individual subject. There is perhaps precious little time to devote to internet-based instructional material. Typing key words (e.g., pharmacology or drug names) in a Web browser generally results in displaying numerous sites of interest. In addition to subject matter interest, there are many general medical sites (e.g., www.medscape.com) providing current articles on a variety of medical practice areas.

NOTE: The Department of Pharmacology would like the students to be aware that the suggested alternative sources of information are intended only to provide additional tools with which to learn pharmacology. They are not intended to supersede information given during lectures or otherwise available in primary pharmacology text sources.

THE PARADOX

A Poem for Students that reflects the timeless, continuing dilemma of how best to teach: (Written by Dr. LeRoy Kuehl, a professor of biochemistry at the University of Utah School of Medicine, published in Whitman, *Creative Medical Teaching*, University of Utah School of Medicine, Department of Family and Preventive Medicine, 1990, pp 92 – 94).

**Three students on the door did knock;
With their professor they would talk.
He welcomed them most cordially;
Asked what could their problem be?
And in the office of that Saint,
They laid before him their complaint:**

**Your course falls short, you can't deny,
Since lecture notes you don't supply.
So all we do in your class,
Is take dictation, notes amass.
Preoccupied with pen and ink,
We have no time to learn nor think.
Your pearls of wisdom, priceless quotes
Slip by while we sit taking notes.**

**The good professor promised this,
That they'd have lecture notes forthwith.
Six days he toiled and evenings too,
And when his task was finally through
Then all his lectures without fail
Were written up in great detail,
And basic concepts were defined,
Because they were all underlined.**

**Next year came students as before
To knock upon that good man's door;
To tell him how he was remiss,
And now their grievance read like this:
In lecture, everything you say
Is in the course notes anyway.
And since we all know how to read,
Of hearing you there is little need.
We could as well remain at home
And learn the subject on our own.**

**Could you not tell us something new,
That isn't in the handout too?
The good professor promised this:
To remedy the flaw forthwith.
Three days he toiled and evenings too,
And when his task was finally through,
He added data and more quotes,
And minor points and anecdotes
To supplement and underscore
The topics that he'd taught before.**

**Next year the students came to bleat:
His lecture notes were incomplete.
For much of what he did expound,
Could nowhere in the notes be found.
Although this caused him much chagrin,
So to their braying he gave in,
And once again without delay
He wrote down all that he would say.**

**And to the old notes this was added-
His outline now was quite padded.**

**When yet another year had passed
And he thought he'd found peace at last,
There welled a cry from student throats
That all he did was read his notes.
The poor professor paced the hall
Feeling like a ping-pong ball.**

**Thus year by year his lectures grew,
And longer were his outlines too.
And students found to their distress
They had to master an excess
Of obscure facts, minute detail
Of information dry and stale.
A situation that attained
All because they had complained.**

**The moral is easy to observe:
Students get what they deserve.**

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Course Design, Organization and Policies

A. What is Pharmacology?

Pharmacology is the study of the effects of chemicals (primarily drugs but also various non-drug substances) on biological systems. As a basic science, pharmacology encompasses a wide variety of knowledge including chemical properties, mechanisms of action, and disposition of chemicals in living systems. Pharmacotherapy is the knowledge applied to the use of drugs in diagnosis, prevention, and treatment of disease.

Drug therapy is the most widely used modality of medical treatment. To provide rational therapy in patient care, the physician must have knowledge of the scientific as well as the therapeutic aspects of drug action. When making a diagnosis or treating a disease, the physician must also be aware of the toxicologic properties of drugs and chemicals to which patients may be exposed. **Understanding the pathology affecting the patient is essential to understanding the pharmacological tools available for treatment of the disease.**

Medical school courses most germane to understanding pharmacology are biochemistry, neurobiology and physiology. Students should have successfully completed these courses prior to enrolling in pharmacology. Microbiology, Pathology and Pathophysiology are excellent co-requisite courses, as the clinical aspects of pharmacology involve the use of drug products in the treatment of disease states. Chemistry, biodynamics and biokinetics are working components of pharmacology, and the science of pharmacology utilizes such knowledge in examining how chemicals affect biological systems. Undergraduate courses in organic chemistry and basic mathematics are essential foundations to the understanding of pharmacology.

B. Course Design and Objectives

This course is designed primarily to introduce students to Pharmacology as a basic science in the practice of medicine. It includes approximately 120 hours of scheduled class time, review and examination sessions. This course provides a balance of instruction interrelating the science of pharmacology with pharmaco-therapeutics. Scientific information, therapeutic effects, clinical usage, mechanism(s) of drug action, side effects, and toxicity are discussed for each class of drugs presented in the course.

The course begins with a foundation of basic principles of drug action. Basic principles apply to all areas covered in the course and are continually re-emphasized, where applicable, to provide a better understanding of clinical uses of drugs. The substantive topics in pharmacology generally focus on mechanisms of action, side effects, toxic effects, approved clinical uses, drug-drug interactions, contraindications to usage, and special clinical applications of drug products.

This syllabus contains specific learning-focused **lecture objectives**. **Read these objectives prior to attending class.** Test questions are based upon these objectives. Mastering the learning objectives will provide you with a sound background for success in national boards and medical practice. To the extent practicable, instruction will include case-based/applied pharmacology. Several sessions are devoted to clinical correlations lectures or problem-solving exercises.

C. Testing and Evaluation of Performance

Performance in the course is evaluated based on seven (7) inter-sessional examinations and a final, the USMLE subject area (mini board) examination. Please refer to the course schedule for dates, times and content coverage of exams.

The materials for these exams will be taken primarily from lecture handouts, all formal presentations, the required textbook, and specific reading assignments. Exam content is cumulative from the beginning of the course throughout the two-semester period of instruction. **Exam 4 (December exam block) and Exam 7 (April exam block) will contain up to 40% of their questions from content covered in exams 1-3 and 4-6, respectively.** To the extent practicable, exams will be styled in the current USMLE format. There is no specific formula for proportioning exam questions along the lines of lecture hours per topic, although an approximate number of questions from each faculty member, proportional to their lectures in an exam block, can be expected. To provide a realistic perspective, exam question content is often integrated between and among various topics in pharmacology and may appear in the form of case-based questions.

Exam valuation and administration:

Exams = 80% of course grade (approximately 11.4% for each of 7 exams).

Mini board = 20% of course grade.

Bonus Points: maximum for course = 5 (5 percentage points from NBME exam at the end of the course, as described below).

The Department of Pharmacology and Toxicology must abide by institutional rules and policies. To this extent, **all exams are mandatory** and missed examinations may be made up only during assigned periods, with the express authorization of the Associate Dean of Student Affairs. The Department is not free to negotiate the consequences of missed exams, regardless of the underlying reason(s). **There are no dropped scores.**

Exams will generally contain 50 to 100 questions given at intervals set by the institutional academic calendar. Exams are timed (range 90 to 120 minutes).

Exam construction, style and environment will simulate, to the greatest extent possible, that of the USMLE Step examinations. Exams will be administered on designated dates and times, as published in the course schedule, and monitored by responsible institutional personnel. Questions will not be allowed during exam sessions as this creates a distraction to others. Students may use the note taking feature in Exemplify to write questions about exam items or problems perceived with the exam. The faculty will evaluate these comments at the time the exams are corrected and take appropriate action. Students must adhere to MSM rules of conduct during examinations as published in the student handbook or distributed otherwise.

Individual examination scores will be posted by the confidential identifier assigned to individual students on Canvas. Students may review examinations with faculty by appointment, preferably within the two weeks following receipt of examination results. Reviewing your examinations in conference with a faculty member provides the best method of addressing subject matter questions. Instructors will provide review questions and test exercises to facilitate your examination preparation.

Course grades are determined according to the composite percent scores as follows:

H (Honors) = 88% - 100%

P (Pass) = 70% - 87%

F (Fail) = <70%

LATE EXAM POLICY

Students are expected to arrive at all examinations 15 minutes before the scheduled start time and be ready to begin the examination five minutes before the scheduled start time. Students commencing NBME subject exams after the announced start time in the room or Exemplify examination 2 or more minutes after the stated start time will be considered to be late. No additional time will be allowed to make up for a late start.

*For the **first episode** of late (on any exam) by a student in an academic year there will be 5% penalty (reduction in grade) for that examination.*

*For the **second late episode** in a year, there will be a 10% penalty for that and ALL REMAINING EXAMS for the year. In addition, two late arrivals will result in a professionalism referral to the Office of Student Affairs and Admissions.*

*For the **third late episode**, penalty will rise to 20% for all remaining exams of the year.*

D. The Mini Board Examination

The most significant opportunity to improve your course performance is your performance on the mini board. **An excellent performance on the mini board can potentially increase your final course average by up to five points.**

One (1) extra bonus percentage point between the 50th and 64th percentile ranking.

Two (2) extra bonus percentage points between the 65th and 74th percentile.

Three (3) extra bonus percentage points between the 75th and 84th percentile.

Four (4) extra bonus percentage points between the 85th and 94th percentile.

Five (5) extra bonus percentage points at the 95th percentile and above.

E. Tutorial Assistance and Course Remediation Policy

Students are encouraged to seek advice and help from the faculty whenever they choose, or if advised to do so. Institution student tutors are available to assist students taking the course. Tutors can be accessed through the Office of Student Learning and Educational Resources (OSLER). You are welcome to come to the department to discuss any matter pertaining to the subject areas or other concerns regarding the course at any time you have free in your schedule. If necessary, pharmacology faculty will meet with you after hours. Please give the faculty member(s) advanced notice so that they can arrange to meet with you. All faculty may be reached directly on their office phone numbers, at the email addresses or in person in their offices as listed on page 5 of this syllabus. Professors will provide study questions or review test questions for their respective sessions. Students are advised to use these practice test questions as a barometer for determining mastery of the material and **NOT** as a tool to memorize what may be tested.

For students who are eligible for remediation of pharmacology, the course director will work with Undergraduate Medical Education (UME) leadership and Pharmacology faculty to align the remediation experience to address the student's learning needs. This is likely to be focal or global review (by the student) of posted learning materials (recorded lectures, lecture slides, textbook

readings, and learning objectives) aligned with the student's needs, faculty or course director guidance and coaching, and scheduled testing which may include in-house block or summative exams and/or the NBME pharmacology subject examination (mini board). Passing the examination as scheduled is required to pass the remediation. Students are reminded, however, that a decision allowing repeat of a course or other means of remediation must derive from deliberations of SAPP. Successful remediation will be reflected in the students' transcripts by entry of a grade of "P".

F. Disability Assistance

Morehouse School of Medicine seeks to provide students with disabilities, equal opportunities and equal access to academic programs, services, and activities. If you have a disability for which you wish to request classroom and testing accommodations, please contact the Office of Inclusive Learning & Accessibility Services at the beginning of each academic year (or as soon as practical). The Office of Inclusive Learning & Accessibility Services is located in the Education MBU. You may also contact the office at 404-756-5778 or via e-mail at OILAS@msm.edu. If the Office of Inclusive Learning & Accessibility Services offers an accommodation, it is the student's responsibility to provide this information to the Course Director.

TIPS TO LEARNING PHARMACOLOGY

Where to Start

Outline of the flow of information and major focus points.....

You cannot possibly learn everything about every drug available....

Remember that you will retain only a fraction of what you study the first time you are exposed to a new subject. While many students can memorize an incredible amount of useful (and useless) information, you must try to organize the material in such a way that you memorize essential facts that may be used as hooks to recall other material. You need to get the most bang for your buck, or most facts learned for each hour of study. Usually, this means grouping drugs and making appropriate associations. Our ultimate goal is not only to teach the facts but to provide a rational thought process by which you can learn to utilize drugs as therapeutic agents.

New drugs will be introduced during your lifetime and even during the course, so develop a flexible framework for handling drug information.....

The best approach is to learn the drugs by class. If you know the characteristics of angiotensin-converting enzyme inhibitors, then when a new one is marketed you will have a framework for comparison.

Many students try to memorize everything about a drug and end up remembering the most trivial facts and forgetting the most important. Often, from your perspective, it is very difficult to know what is most important and what can be omitted to make room for essential facts. This is a result of the way most textbooks are organized. They give general information about the pathophysiological condition being treated and/or the drug class and then provide details about each individual agent in the class. This is a good way to be thorough, and it is very useful when you need to go back and look up a detail about a drug. However, it is not the best way for the beginning student to learn the information.

To help you decide what is the most important information, you should develop a drug trivia sorter (see example below). The trivia sorter will help you to manage voluminous information by knowing what to include and what to leave out. Apply this scheme to prototype drugs in each of the many drug classes you study. Soon, the 7 recommended **Must Know** items become automatic steps in the study habit of learning pharmacology.

Drug Trivia Sorter

You Must Know:

1. **Drug Class and Generic Name:** Learn the class of drugs (usually based on the mechanism of action and/or clinical usage; sometimes based on chemical classification) and the prototype drug(s) of each class of drugs.
2. **Pharmacodynamics:** Learn “how/where drugs act on the body,” the site (location) of action – the receptor, organ – mechanism of action. From this information, you can often predict the clinical use(s) of drugs.
3. **Pharmacokinetics:** Learn “how the body acts on drugs.” How is the drug administered, where does it go, what changes to the drug structure take place, and how is it eliminated?
4. **Adverse Effects:** “Unwanted effects” – what are the toxic, sometimes life-threatening, or teratogenic (birth defects) actions of the drug?
5. **Clinical Uses:** Learn the drug’s indications for therapy – this is often predictable by the drug’s pharmacodynamic properties.
6. **Contraindications:** What are the circumstances under which the drug cannot be used or must be used with caution?
7. **Drug Interactions:** Learn the drug-drug, drug-food, drug-environmental chemical interactions that result in the loss of therapeutic benefit or the enhancement of toxicity.

The *DRUGS OF CHOICE* are often very important to know; however, these are subject to change. Therefore, you should confirm that the drug in question is the drug of choice at the hospital or clinic where you are located. *FATAL ADVERSE EFFECTS*, even if rare, are important to know for your patients’ sake, make a note to learn them.

Drug Name Recognition: *Tongue twisting chemical language.....*

The approach to name recognition is a bit different. This comes about from the past experience of medical students, who sat down to take an exam in pharmacology, then got to a question that contained a drug name that was unfamiliar. Often the question was missed, but if the drug class could have been identified, then answering the question usually becomes less problematic. Students who are having trouble with name recognition should make flash cards (or lists) with only the drug name on one side and the drug class on the other side. Skip the easy ones. Quiz yourself during breakfast or during breaks between classes. As you learn the drugs, take them off your list or remove the card from the stack. Occasionally add back in all the names. If you only get a few more questions right on your exam or the boards or have to look up one less drug after rounds, the few minutes that this takes will have been worth it.

Since students are only expected to know generic names of drugs for purposes of taking the USMLE, primarily generic names will be used throughout this course. Trade names may be included in the handouts and textbook. Hopefully, these suggestions will provide a more organized approach to learning the basic information about drugs and help to develop a working vocabulary of drug names.

The language of pharmacology (drug names) derives largely from some form of abbreviated chemical name referred to as a "generic" drug name. Drug cards provide a rapid method of assessing memory of categorization and learning a few facts about the drug. Writing your own drug cards can be a valuable way to learn. The amount of information put on drug cards should be limited so as not to make the cards too time consuming to review. Drug cards should be designed as small doses of information to be taken regularly to allow accumulation of a tolerable amount of knowledge rather than an acute overdose of material.

Select key (prototype) drugs to card. Making a card for every drug encountered in the course would be impractical. Drug therapy is an applied science, and pharmacology cannot be learned in a vacuum. Before studying pharmacology, you will have been exposed to the actions of drugs presented in other basic science courses, especially biochemistry, physiology, pathophysiology and microbiology.

Drug actions in other basic science courses are generally not presented in the same manner or depth of discussion as in pharmacology, but the point being made here is the importance of INTEGRATION OF SCIENTIFIC AND MEDICAL KNOWLEDGE as it relates to the entire process of the education of a physician. To this end, the study of pharmacology requires review of other basic science disciplines. This is a good opportunity for students to take advantage of and maximize their studying time in reviewing for USMLE Step I.

Many of the changes in the course design and delivery are based upon input from students who have taken the course. In this regard, student input is welcomed at any time during the course and not just on evaluations at the close of the course.

Note: The Course Director reserves the right to adjust the course syllabus whenever necessary.

Knowledge Objectives in Pharmacology

Function & Purpose of Knowledge Objectives

The broadest instructional objective of the pharmacology course is to present to the student a system of instruction and guidance for learning basic science, clinical concepts and other information regarding drug action. Instructional learning objectives have been written to assist students in focusing on specific aspects of the subject matter. Certainly, these objectives do not represent everything a student (or physician) may learn in the classroom or through professional practice. The knowledge objectives represent a starting focus of a life-long process of learning subject matter pertinent to the practice of medicine.

Assessment of student progress in learning objectives may be measured in a variety of ways. On written examinations, students are generally asked to identify information, apply learned material to a given situation and sometimes, to rationalize an outcome given certain facts in a situation or case. Questions asked of a student in class elicit whether the student has made adequate pre-class preparation, the level of comprehension and progress and how well the student thinks in extemporaneous situations requiring spontaneous conversation to be heard by an audience. Knowledge objectives provide guidance to the student in the process of learning facts and concepts essential for understanding a course of instruction. Knowledge objectives provide a means of organizing and focusing a learning experience as well as setting expectations of learning outcomes.

Drug Lists: *How many of these drugs do I need to know.....*

Invariably, knowledge objectives in pharmacology derive from specific drugs or classes of drugs. Representative examples of prototype and other key drugs are included in the syllabus to provide an initial focus of study. The text provides an exhaustive compendium of drugs. Specific drug lists for each topic will be provided as part of classroom instruction. It is impractical to provide drug lists in advance of instruction as drugs are continually removed from or added to the market. Drug lists provided in class allow for flexibility necessary to convey the most current information. Individual instructors will delineate which drugs you will be responsible to learn.

The faculty may choose to include knowledge objectives and drugs from lists published by national professional associations.

Learning Objectives/Lecture Session Activities

Note: Unless otherwise indicated, text reading assignments refer to Katzung's Basic & Clinical Pharmacology, by Todd W. Vanderah, 16TH Ed., 2024, McGraw Hill, NY. An electronic version of the text is available in Access Medicine from MSM's Library of electronic textbooks: (Available from Access Medicine at [Katzung's Basic & Clinical Pharmacology, 16th Edition | AccessMedicine | McGraw Hill Medical \(mhmedical.com\)](https://mhmedical.com))

Session 1

General Principles I Pharmacodynamics: Receptors and Drug Receptor Interactions Text Assignment: Chapter 2

Learning Objectives:

1. Define terms related to drug action:

Dose-response relationship

Agonist Antagonist

Potency Efficacy

Therapeutic Index Tolerance/Tachyphylaxis

Synergism

2. Discuss properties of drug receptors relative to the concepts of:

Specificity of action/Dose-response relationships

Concepts and nature of antagonism:

Partial Agonists

Competitive Antagonism

Noncompetitive Antagonism

Pharmacokinetic Antagonism

Physiologic Antagonism

Chemical Antagonism

3. Relate specific examples of drug-receptor interactions and consequences relative to the following receptor types:

Nicotinic

Muscarinic

Adrenergic

Gabaminergic

Sessions 2 & 3

General Principles II & III: Pharmacokinetics Chemical Characteristics of Drugs Drug Absorption, Distribution and Elimination

Text Assignment: Chapters 1 and 3

Learning Objectives:

1. Identify factors that determine dosage and duration of drugs required to achieve therapeutic effects.
 - a. Explain the concept of apparent volume of distribution (V_d), how it is determined, and how V_d is affected by the drug chemistry.
 - b. Explain the concept of clearance (Cl).
 - c. Explain the difference between first order and zero order elimination of drugs.
 - d. Explain the relationship between V_d , clearance (Cl), and half-life ($t_{1/2}$).
 - e. Determine V_d , Cl and $t_{1/2}$ from graphical data depicting drug concentration over time.
 - f. Explain how $t_{1/2}$, steady-state drug concentrations and dosing frequency affect drug accumulation.
2. Explain bioavailability and its relationship to routes of administration.
 - a. Discuss the concept of first-pass elimination and how it affects bioavailability.
 - b. Describe the role of protein and tissue binding in drug availability and effects on V_d , Cl & $t_{1/2}$ and therefore dosing.
3. Explain how V_d , Cl and $t_{1/2}$ relate to adjusting doses or dosage schedules.
 - a. Describe how to determine dosing rates to achieve and maintain target concentrations.
 - b. Describe how dose levels and dosing frequency interact to maintain target concentrations.
 - c. Explain the concept of maintenance dose and how this is used.
 - d. Explain the concept of loading dose and how this is used.
 - e. Explain how the sustained effect of drugs even after drug concentrations have decreased affects frequency of dosing
4. Discuss the concept of drug ionization as defined by the Henderson-Hasselbalch equation and its role in drug absorption/distribution/elimination.
 - a. Explain the concept of weak acid, weak base, and pKa in terms of drug protonation and lipid solubility.
 - b. Explain the rationale for acidifying or alkalinizing the urine to enhance or slow excretion of drugs that are weak acids and weak bases.
 - c. Describe factors that affect renal excretion of drugs.
 - d. Describe factors that affect hepatic excretion of drugs.
5. Describe the role of transporters in distribution of drugs.

Session 4

General Principles IV: Pharmacokinetics Drug Metabolism and Excretion

Text Assignment: Chapters 3 and 4

Learning Objectives:

1. Indicate the key factors that influence the biotransformation of drugs.
2. Describe the role of enzyme-mediated metabolism for drug excretion.
3. Describe the difference in types of reactions between Phase 1 and Phase 2 enzymes.
4. Identify key sites of drug metabolism.
5. Describe the principle metabolic changes in drugs accomplished by Phase I biotransformation reactions.
 - a. Identify the relative importance of the cytochrome P450 families.
 - b. Describe the functions of the cytochrome P450 enzyme system.
6. Describe the principle metabolic changes in drugs accomplished by Phase II biotransformation reactions.
 - a. Describe the effect of conjugation in Phase II reactions.
 - b. Using acetaminophen as an example, illustrate the interplay between Phase I and Phase II biotransformation reactions.
7. Describe factors affecting enzyme expression in drug metabolism.
 - a. Explain the mechanisms of enzyme induction and enzyme inhibition in terms of drug substrates.
 - b. Give examples of pharmacogenetic influences on drug metabolism.
8. Describe other factors (physiological, pharmacological, pathological, nutritional) that may alter drug metabolism.
9. Explain the primary factors involved in drug interactions.
10. Summarize the primary cause of variation in drug metabolism.

Session 5

Introduction to Pharmacogenomics Text Assignment: Chapter 5

1. Identify factors that can impact a patient's response to pharmacotherapy.
2. Define the following aspects of the human genome:

Allele	Heterozygous	Polymorphism or variant
Allele frequency	Homozygous	PM, IM, EM, or UM
Coding single nucleotide polymorphisms (cSNPs)	Insertion/deletion (indel)	Promoter
Copy number variations (CNVs)	Intron	Protein
Exon	Linkage disequilibrium	Stop codon
Gene	Mutation	Synonymous SNPs
Haplotype	Noncoding region polymorphism	Transcript
Hardy Weinberg equilibrium	Nonsynonymous SNPs (nsSNPs)	Wild type

3. Define the following genetic epidemiology terms:

Association	Linkage
Complex trait	Race
Ethnicity	Simple (Mendelian) trait
Genetic ancestry	Whole exome sequencing (WES)
Genome-wide association studies (GWAS)	Whole genome sequencing (WGS)

4. Provide a brief overview of the history of genomics research.
5. Differentiate between pharmacogenetics and pharmacogenomics.
6. Demonstrate an awareness of the use and impact of pharmacogenomics in precision medicine and within the health care system.
7. Navigate select pharmacogenomics-related databases and apply the knowledge gained in a case-based clinical setting.
8. Effectively communicate pharmacogenomics-related pharmacotherapy and drug information using relevant pharmacogenomics-related databases.
9. Provide established examples of pharmacogenomic variants in CYP and phase II drug metabolizing enzymes (DMEs) that influence drug therapy.
10. Describe the potential clinical implications of DME genetic polymorphism on drug disposition.
11. Discuss the roles that plasma membrane transporters (PMTs) play in drug disposition and action.
12. Describe how PMTs influence drug action in the intestine, liver, kidney, and blood-brain barrier (BBB).
13. Cite specific examples of how disease, pharmacogenomic variation and population differences may affect PK and PD of substrate compounds for DMEs and PMTs.
14. Provide an overview of select "omics" research disciplines:

Bioinformatics
Epigenomics
Metabolomics
Metagenomics
Proteomics

Detail future directions and challenges of pharmacogenomics in the era of precision medicine.

Session 6

Topic: Introduction to Autonomic Pharmacology Text Assignment: Chapter 6

Learning Objectives:

1. Illustrate, by means of a diagram, the major anatomical features of autonomic and somatic motor nerves.
2. Explain, by the diagram created above, how drugs having action at: a) cholinergic receptors affect sympathetic nerve activity and b) adrenergic receptors affect cholinergic nerve activity.
3. Outline the sequential steps involved in neurochemical transmission in somatic motor nerves and autonomic nerves.
4. Describe, in diagrammatic form, the major sites and processes whereby drugs can modify function of somatic and autonomic (cholinergic/adrenergic) nerves.
5. Describe how the baroreceptor reflex alters parasympathetic and sympathetic modulation of cardiovascular function.
6. Outline the essential steps involved in the disposition and catabolism of acetylcholine, norepinephrine and epinephrine.
7. Describe, in diagrammatic form, the major sites and processes involved in the synthesis of acetylcholine, dopamine, norepinephrine and epinephrine.
8. Compare the sites and contrast the actions of acetylcholine on the two major cholinergic receptor types: muscarinic and nicotinic.
9. Compare and contrast the actions of norepinephrine, epinephrine, dopamine, and isoproterenol on the major adrenoceptor types: adrenergic (α_1 , α_2 , β_1 , β_2 , β_3) and dopaminergic (DA_1 , DA_2).
10. Describe the cellular signaling mechanisms that transverse the action of catecholamines and acetylcholine on receptors in various tissues.
11. Explain the concept of autoreceptors and heteroreceptors in the regulation of neurotransmitter release.
12. Explain the concept of denervation supersensitivity and how drugs may be used to localize the site(s) of lesion in the autonomic nervous system (ANS).
13. Explain the concepts of tachyphylaxis, up-regulation and down-regulation as related to therapy with autonomically active drugs.

Session 7

Cholinergic Pharmacology I - Cholinergic Receptor Agonists (Direct- and Indirect-Acting) Text Assignment: Chapter 7

Learning Objectives:

1. Compare the structure/activity relationships of methacholine, carbachol, bethanechol, and pilocarpine to acetylcholine with respect to: a) the balance of muscarinic/nicotinic activity and b) degradation by acetylcholinesterase.
2. Compare the tissue sites and cellular transduction mechanisms of the various muscarinic and nicotinic receptor subtypes.
3. Define the role of nitric oxide in the cellular action of muscarinic receptor agonists.
4. Identify the effects of cholinergic agonists on organ systems including the cardiovascular system, GI and GU systems, eye, respiratory system, exocrine glands and brain.
5. List and explain the differences among the principal types of indirect-acting cholinergic agents.
6. Illustrate the principle of acetylcholinesterase regeneration following organophosphate toxicity and explain the mechanism of action and role of oxime drugs.
7. Outline the critical steps in treating insecticide (organophosphate) toxicity.
8. Explain the mechanism(s) of action and rationale for the use of direct and indirect-acting cholinergic agonists in the management of glaucoma, myasthenia gravis, GI/GU dysfunction, cardiovascular dysfunction and reversal of neuromuscular blockade induced during surgery.
9. Describe how edrophonium is used to distinguish between myasthenic weakness and cholinergic crisis.

Representative Drug Agents: Cholinergic Receptor Agonists

Cholinergic Agonists (Direct)

1. acetylcholine
2. bethanechol (Urecholine)
3. carbachol
4. pilocarpine
5. nicotine (Habitrol, Nicoderm)

Cholinesterase Inhibitors (Indirect)

1. physostigmine (eserine)/neostigmine (Prostigmin)/pyridostigmine (Mestinon)/edrophonium (Tensilon)
2. DFP (diisopropylfluoridate)/parathion (Etilon)/echothiophate (Phospholine)
3. pralidoxime (2-PAM, Protopam)

Session 8

Cholinergic Pharmacology II - Cholinergic Receptor Antagonists (Including NJM Blocking Drugs and Ganglionic Blocking Drugs) & Skeletal Muscle Relaxants

Text Assignment: Chapters 8 and 27

Learning Objectives:

1. Describe the mechanism of action of muscarinic receptor antagonists in terms of the prototype agent - atropine.
2. Identify the subclasses of muscarinic receptors as defined by the actions of atropine and pirenzepine.
3. List the major effects of antimuscarinic agents on the function of the following organs/organ systems: CNS, GI, CV, eye, GU, lungs, sweat glands.
4. Explain the clinical uses and side effects of antimuscarinic drugs relative to their effects on specific organs and/or organ systems.
5. Define conceptually why benztropine would be more appropriate in treating symptoms of Parkinsonism than ipratropium.
6. Describe the role of atropine in treating cholinergic toxicity resulting from a cholinomimetic drug.
7. Identify hallmark signs and symptoms of atropine toxicity and suggest appropriate management.
8. Determine the pathophysiological basis of the contraindications to the use of antimuscarinic drugs.
9. Describe the mechanism(s) of action of ganglionic blocking drugs. Identify two types of ganglionic blocking agents based upon mechanism of action.
10. Use an autonomic organ systems chart, and/or the Hexamethonium Man model to predict the effects of ganglionic blockade.
11. Specify the limited clinical use(s) of ganglionic blocking drugs and why their use is limited.
12. List the typical side effects of ganglionic blocking drugs and provide an explanation based upon the division of the ANS that is affected.
13. Differentiate between the mechanisms of action of the major classes of (nondepolarizing versus depolarizing) neuromuscular blocking agents.
14. Identify the key pharmacokinetic differences among the depolarizing and nondepolarizing neuromuscular blocking drugs.
15. List the neuromuscular blocking drugs currently in use and construct a chart that illustrates their effects on autonomic ganglionic transmission (blood pressure), cardiac muscarinic receptor function (heart rate) and release of histamine from mast cells.

16. Provide a rationale for reversal of curare like drugs by acetylcholinesterase inhibitors.
17. Describe the considerations involved in the use of neuromuscular blocking drugs in patients with neuromuscular disease, infection, burns or genetic alterations in pseudocholinesterase.
18. Suggest why succinylcholine can induce hyperkalemia in patients suffering from conditions involving denervated muscle.
19. List the potential drug interactions that may arise as a result of concurrent use of neuromuscular blockers with anesthetics, antibiotics, and local anesthetics/antiarrhythmics.
20. Identify drugs used to treat conditions involving muscle spasm (including malignant hyperthermia) and describe their mechanism and sites of action; describe the clinical indications for the specific agents listed.

Representative Drug Agents: Cholinergic Antagonists/Skeletal Muscle Relaxants

Cholinergic (Muscarinic) Antagonists

1. atropine
2. scopolamine (Transderm-Scop)
3. methantheline (Banthine)
4. ipratropium (Atrovent)
5. tropicamide (Mydracyl)
6. pirenzepine
7. oxybutynin (Ditropin)
8. flavoxate (Urispas)

Cholinergic (nicotinic/Ganglionic) Agonists/Antagonists

1. nicotine
2. hexamethonium (C6)
3. mecamylamine (Inversine)

Neuromuscular Junction/Skeletal Muscle Relaxants

1. botulinus toxin (Oculinum)
2. d-tubocurarine
3. atracurium (Tracrium)
4. succinylcholine (Anectine)
5. dantrolene (Dantrium)
6. cyclobenzaprine (Flexeril)
7. carisoprodol (Soma)

Session 9

Adrenergic Agonists Text Assignment: Chapter 9

Learning Objectives:

1. Outline the major tissue distributions of adrenoceptor subtypes.
2. Define the mechanism meaning of the terms sympathomimetic, relatively selective, indirect-acting, and inverse agonist.
3. List at least one sympathomimetic drug for each mode of action: direct, indirect or mixed (direct & indirect).
4. Classify the major classes and subtypes of adrenergic receptors and compare the activities of norepinephrine, epinephrine, isoproterenol, and dopamine on the various receptor subtypes.
5. Outline the signal transduction mechanisms associated with activation of various types of adrenergic receptors.
6. Define how changes in chemical structure (structure-activity relationship) affect the pharmacologic activity and disposition of adrenergic agonists.
7. Identify the effects of adrenergic agonists on major organ systems, including intermediary metabolism and endocrine function.
8. List and classify sympathomimetic drugs according to their uses in clinical practice.
9. Based upon their receptor profiles, discuss the rationales for the therapeutic uses of adrenoceptor agonists.
10. Identify the major toxic effects and contraindications of sympathomimetic drugs.

Representative Drugs: Adrenergic Agonists

CATECHOLAMINES

1. dopamine (Intropin)
2. epinephrine (Sus-Phrine, Eppy)
3. isoproterenol (Isuprel)
4. norepinephrine;
5. levarterenol (Levophed)

SYMPATHOMIMETIC AMINES

1. clonidine (Catapres)
2. methyldopa (Aldomet)
3. phenylephrine (Neo-synephrine)
4. hydroxyamphetamine (Paredrine)
5. terbutaline (Brethine)
6. ritodrine (Yutopar)
7. dobutamine (Dobutrex)
8. bromocriptine (Parlodel)
9. cocaine
10. phenylpropanolamine

Session 10

Adrenergic Antagonists and Adrenergic Neuronal Suppressants Text Assignment: Chapter 10

Learning Objectives:

1. Classify adrenoceptor antagonists (sympatholytics) according to their relative receptor affinities.
2. Explain the concept demonstrated by an antagonist that can produce the phenomenon referred to as epinephrine reversal.
3. Differentiate among various beta receptor antagonists on the basis of pharmacodynamic (receptor selectivity, partial agonist activity) and pharmacokinetic (solubility characteristics, disposition) characteristics.
4. Differentiate among the alpha adrenoceptor antagonists relative to their pharmacodynamic characteristics: receptor selectivity, type of receptor antagonism, major sites of action.
5. Explain and contrast the major pharmacodynamic actions of alpha and of beta adrenergic receptor antagonists on the major organ systems.
6. Discuss adrenergic receptor antagonists with specific regard to their therapeutic use in cardiovascular disorders, glaucoma, and hyperthyroidism.
7. Identify precautions and contraindications to the use of adrenoceptor antagonists as well as the major adverse effects associated with these drugs.
8. Provide examples and define the sites and mechanisms of action of adrenergic neuronal suppressants.
9. Identify the primary clinical indications for adrenergic neuronal suppressants.
10. List the adverse effects associated with drugs that suppress adrenergic neuron function.

Representative Drugs:

Non-selective α antagonist

1. phentolamine (Regitine)
2. phenoxybenzamine (Dibenzyline)

Selective α_1 -antagonist

1. prazosin (Minipress)
2. terazosin (Hytrin)
4. doxazosin (Cardua)

Selective α_2 -antagonists

1. yohimbine (Yocon)

Non-selective β_1/β_2 - antagonists

1. propranolol (Inderal)
2. nadolol (Corgard)
3. timolol (Timoptic, Blocadren)

Selective β_1 -antagonists

1. atenolol (Tenormin)
2. metoprolol (Visken)

Non-selective β antagonists with α_1 antagonist activity

1. labetalol (Normodyne)
2. carvedilol (Coreg)

Adrenergic Neural Suppressants

1. clonidine (Catapres)
2. methyldopa (Aldomet)
3. reserpine (Serpasil)
4. guanadrel (Hylorel)
5. metyrosine (Demser)
6. brimonidine (Alphagan)

Session 11

Treatment of Inflammation 1: Eicosanoids (Prostaglandins and Related Compounds)

Text Assignment: Chapter 18

Learning Objectives:

1. Describe the basic structure and nomenclature of eicosanoids.
2. Describe the arachidonic acid cascade regarding cyclooxygenase (PGE₁, PGE₂, PGF₂, PGI₂, TXA₂) and lipoxygenase pathways (leukotrienes).
3. List the key enzymes in the overall pathways and identify drugs that affect each enzyme.
4. Discuss the pharmacological actions of eicosanoids on tissues throughout the body
5. Discuss the therapeutic uses, adverse effects, and contraindications to the use of eicosanoids in the following cases.
 - a. bronchodilation
 - b. maintenance of patent ductus arteriosus
 - c. cytoprotective effect in ulcers (misoprostol)
 - d. relieving symptoms of dysmenorrhea
 - e. glaucoma

Representative Drugs:

1. alprostadil
2. carboprost (Prostin F₂ α)
3. dinoprost (Prostin F₂ α)
4. dinoprostone (PGE₂, Prostin E₂)
5. latanoprost
6. misoprostol
7. montelukast
8. zafirlukast
9. zilueton

Session 12

Treatment of Inflammation 2: Nonsteroidal Anti-inflammatory Drugs (NSAIDs); Acetaminophen Text Assignment: Chapter 36

Learning Objectives:

1. Review eicosanoid (prostaglandins, etc.) biosynthesis.
2. Discuss the pharmacology of salicylates (possible mechanism(s) and sites of action of nonsteroidal anti-inflammatory drugs with respect to their analgesic, antipyretic, and anti-inflammatory actions).
3. Discuss dosage variations of aspirin relative to its antipyretic, analgesic, anti-inflammatory, and platelet inhibition actions.
4. Discuss the metabolism of acetaminophen relative to the role of cytochrome P450 and the mechanism of acetaminophen toxicity.
5. Discuss the pharmacology of select cyclooxygenase I and II (COX I & II) inhibitors (salicylates and nonsalicylates), differentiating selective inhibitors of COX II.
6. Discuss the adverse effects, toxicities, and contraindications to NSAID use.

Representative Drugs:

A. Nonsteroidal Anti-inflammatory Drugs (NSAIDs):

1. aspirin
2. celecoxib
3. diflunisal
4. ibuprofen
5. indomethacin
6. naproxen

B. Acetaminophen

Session 13

Autacoids

Text Assignment: Chapter 16

Learning Objectives:

1. Discuss the major organ system effects of histamine and serotonin
2. Identify the signal transduction pathways involved with activation of histamine receptors.
3. Identify the histamine receptor(s) involved in smooth muscle contraction and gastric mucosa secretions.
4. Explain why blockade of allergic responses is only partially successful with histamine antagonists.
5. Discriminate between the H₁-receptor mediated effects of classical antihistamines and their actions on other types of receptors.
 - a) Contrast the sedative action of the different classes of H₁-blockers.
 - b) Compare efficacy and side effects of H₁ - antagonists.
6. Describe the pharmacology of H₂ antihistamines; identify the four members of this group.
7. Describe the principal actions of 5-hydroxytryptamine on the heart, vasculature and gastrointestinal tract.

Representative Drugs:

H₁-histamine - antagonists	Diphenhydramine [Benadryl]	Loratadine [Claritin]
	Tripeleennamine [Pelamine]	Cetirizine [Zytrec]
	Cyclizine [Marezine]	Fexofenadine [Allegra]
	Chlorpheniramine [Chlor-Trimeton]	Promethazine [Phenergan]
5-HT antagonists -	Ondansetron [Zofran]	Cyproheptadine [Periactin]
Tricyclic antidepressants -	Amitriptyline [Elavil]	
	Nortriptyline [Pamelor]	
Monoamine oxidase inhibitors -	Phenelzine [Nardil]	Isocarboxazid [Marplan]

Session 14

Immunopharmacology Text Assignment: Chapter 55

Learning Objectives:

1. Contrast the use of therapeutic agents for cancer chemotherapy versus immunosuppression, in terms of dosage and degree of selective toxicity.
2. For the following classes of immunosuppressant drugs:
 - antibodies
 - glucocorticoids
 - calcineurin inhibitors
 - proliferation signal inhibitors
 - cytotoxic agents
 - a. Identify the target for their immunosuppressive actions.
 - b. Discuss their mechanism(s) of action in immunosuppression.
 - c. Distinguish between their relative selectivity for immune cells versus non-target cells.
3. Discuss the mechanisms of action of the cytokine agents as biological response modifiers:
 - a. interleukins
 - b. colony stimulating factors

Representative Drugs

Immunosuppressants

Rh₀(D) Immune globulin [Rhogam]
rituximab [Rituxan]
muromonab - CD3 [Orthoclone]
basiliximab [Simulect]
adalimumab [Remicade]
prednisone
cyclosporine [Sandimmune] tacrolimus [Prograf]
sirolimus [Rapamune]
cyclophosphamide [Cytosan]
methotrexate [Folex]
azathioprine [Imuran]
mycophenolate mofetil [Cellcept]

Immunostimulants

interleukin-2 [Aldesleukin, Proleukin]
filgrastim [Neupogen] sargramostim [Leukine] oprelvekin [Neumega]

Sessions 15 & 16

Cancer Chemotherapy Text Assignment: Chapter 54

Learning Objectives:

1. Explain the difference between primary resistance and acquired resistance of cancer cells to chemotherapy.
2. Discuss the rationale for treating cancer with combination chemotherapy
3. Explain the difference between Cell-Cycle Specific (CCS) and Cell-Cycle Non-Specific (CCNS) drugs in terms of their mechanisms of action as anti-neoplastic agents.
4. For the following classes of cancer chemotherapeutic drugs:

alkylating agents

antimetabolites

alkaloids

antibiotics

hormonally active agents

growth factor/signal inhibitors

- a. Identify the target for their actions.
- b. Discuss their mechanism(s) of action.
- c. Mechanisms of acquired drug resistance.
- d. Sites and severity of both acute and delayed drug toxicities.

Representative Drugs

1. Alkylating agents

cyclophosphamide [Cytosan] mechlorethamine [Mustargen]

carmustine (BCNU)

procarbazine [Matulane] cisplatin [Platinol] carboplatin [Paraplatin]

2. Antimetabolites

methotrexate (MTX) [Folex, Mexate]

fluorouracil (FU) [Efudex] floxuridine (5-fluorodeoxyuridine) [FUDR]

cytarabine, Ara-C [Cytosar-U]

mercaptopurine [Purinethol] thioguanine

3. Mitotic inhibitors

vinblastine [Velban]
paclitaxel [Taxol]

vincristine [Oncovin]
docetaxel [Taxotere]

vinorelbine [Navelbine]

4. Topoisomerase inhibitors

irinotecan [Camptosar]
etoposide [VePesid]

topotecan [Hycamtin]
teniposide [Vumon]

5. Antibiotics

doxorubicin [Adriamycin]

daunorubicin [Cerubidine]

bleomycin [Blenoxane]

6. Hormonal agents

tamoxifen [Nolvadex]
fulvestrant [Faslodex]

toremifene [Fareston]

aminoglutethimide [Cytadren]

anastrozole [Arimidex]

letrozole [Femara]

exemestane [Aromasin]

leuprolide [Lupron]

flutamide [Eulexin]

7. Growth factor/signal inhibitors

imatinib mesylate [Gleevec]

gefitinib [Iressa]

trastuzumab [Herceptin]

bevacizumab [Avastin]

tositumomab [Bexxar]

denileukin diftitox [Ontak]

bortezomib [Velcade]

Sessions 17 - 21

Teaching Goals:

1. Focus on: mechanisms of action, mechanisms of resistance, major adverse/toxic effects, differences between and among drugs in the same class, preferred usage, drug interactions, and select indications/spectra of action.
2. Present overview of select drugs used in the treatment of bacterial, superficial and systemic fungal infections, malaria, amebiasis and other protozoal infections, and helminthiasis.

Session 17

Anti-Infectives I: Antibacterial Agents Text Assignment: Chapters 43, 51

1. Distinguish between bactericidal and bacteriostatic agents
2. Explain the mechanisms of action of β -lactam antibiotics, glycopeptides, and other cell wall or cell membrane active agents
3. Describe general mechanisms of resistance and the specific mechanisms for the major agents
4. Explain the role of probenecid in penicillin activity
5. Discuss the principle of β -lactamase inhibitors with penicillins.
6. Explain the pharmacological basis of combining imipenem with cilastatin
7. Describe basic pharmacokinetic properties of the various cell wall synthesis inhibitors
8. Describe the generations of cephalosporins with specific examples, and the differences in their antimicrobial spectrum and pharmacokinetic properties
9. Describe primary therapeutic indications of the agents discussed
10. Describe principal adverse effects of the agents discussed
11. Describe principal contraindications of the agents discussed

Representative drugs will be presented as part of class discussion.

Session 18

Anti-Infectives II: Antibacterial Agents Text Assignment: Chapters 44 & 45

Mechanism of Action

1. Discuss the mechanism of action of each class of protein synthesis inhibitors
2. Explain the mechanism of acquired drug resistance
3. Explain the rational basis for combination therapy with an aminoglycoside and a penicillin, cephalosporin or vancomycin
4. Explain the difference between time-dependent and concentration-dependent killing actions of antibiotics and what is meant by “post-antibiotic effect”.

Pharmacokinetics

1. Describe the pharmacokinetic properties of each class of protein synthesis inhibitors
2. Discuss the need for, and the method of dose adjustment for aminoglycosides in patients with compromised renal function

Adverse Effects & Drug Interactions

1. Discuss the main adverse effects of each class of protein synthesis inhibitors
2. Describe the major drug interactions of tetracyclines, macrolides, chloramphenicol and streptogramins

Therapeutic Uses

Describe the primary therapeutic indications for each class of protein synthesis inhibitors

Special Features

Describe the distinctive features of highlighted drugs in each class

Representative drugs will be presented as part of class discussion.

Session 19

Anti-Infectives III: Antibacterial Agents

Text Assignment: Chapters 46 & 47

Mechanism of Action

1. Describe the mechanism of action of each class of antimicrobial agent presented
2. Describe the first-line antitubercular drugs and explain the differences in their mechanisms of action
3. Explain the mechanism of acquired drug resistance of all agents presented
4. Discuss the synergistic inhibition due to sequential blockade with sulfamethoxazole-trimethoprim

Pharmacokinetics

Describe the pharmacokinetic properties of each class of antimicrobial agent presented

Adverse Effects & Drug Interactions

1. Discuss the main adverse effects of each class of antimicrobial agent presented
2. Explain the drug interactions of sulfonamides with anticoagulants and other drugs
3. Explain the drug interactions of fluoroquinolones
4. Explain the drug interactions of isoniazid
5. Explain the drug interactions of rifampin with anticoagulants and other drugs, such as oral contraceptives

Therapeutic Uses

1. Describe the primary therapeutic indications for each class of antimicrobial agent
2. Describe the regimen recommended for preventive antimycobacterial therapy and for conventional chemotherapy

Representative drugs will be presented as part of class discussion.

Session 20

Anti-Infectives IV: Antifungal Agents Text Assignment: Chapters 48

Mechanism of Action

1. Discuss the mechanism of action of each class of antifungal agent presented
2. Discuss the mechanisms of acquired drug resistance of each class of antifungal agent

Pharmacokinetics

Describe the major pharmacokinetic properties of the various antifungal agents

Adverse Effects & Drug Interactions

1. Describe the important adverse effects of the various antifungal agents
2. Describe the drug interactions of the various antifungal agents

Therapeutic Uses

Describe the major therapeutic indications of the antifungal drugs

Representative drugs will be presented as part of class discussion.

Session 21

Anti-Infectives V: Antiprotozoal and Anthelmintic Agents Text Assignment: Chapters 52 & 53

1. Describe the life cycles of the malaria and amebiasis parasites
2. Describe the mechanism(s) of anti-parasitic action of each drug agent presented relative to actions expressed in parasite versus host cells, and biochemical pathways
3. Discuss the rationale for choosing one anti-parasitic agent versus another
4. Discuss the major therapeutic indications of the antiprotozoal and anthelmintic agents presented
5. Explain the rationale for the use of specific drug combinations
6. Discuss major adverse effects and precautions to use of each drug agent presented in this series

Representative drugs will be presented as part of class discussion.

Session 22:

Anti-Infectives VI

Clinical Correlations

Case focus discussion: Anti-infective Therapy

Session 23

Antihypertensive Drugs Text Assignment: Chapter 11

Teaching Goals:

1. The actions of antihypertensive drugs relative to peripheral and central control of blood pressure.
2. Concepts and designs of the clinical use of antihypertensive drugs.

Learning Objectives:

1. List the major groups of antihypertensive drugs and give examples of drugs in each group.
2. Describe the compensatory responses to each of the major types of antihypertensive drugs.
3. List the major sites of action of sympathoplegic drugs and give examples of drugs that act at each site.
4. List the mechanisms of action of vasodilator drugs
5. List the major antihypertensive vasodilator drugs and describe their effects
6. Describe the differences between the two types of angiotensin antagonists.
7. List the major toxicities of the prototype antihypertensive agents.
8. Explain why some combinations of antihypertensive drugs are rational and appropriate and others are not.

Representative drugs:

1. Diuretics (hydrochlorothiazide, eplerenone)
2. Sympathoplegics
 - a. CNS sympathetic outflow (methyldopa, clonidine)
 - b. Alpha or beta receptors (prazosin, propranolol)
3. Vasodilators
 - a. Oral vasodilators (nitrates, calcium channel blockers, bidil)
 - b. Calcium blockers (nifedipine; verapamil, diltiazem)
 - c. Parenteral vasodilators (hydralazine, minoxidil)
4. Angiotensin antagonists
 - a. ACE inhibitors (captopril)
 - b. Receptor blockers (losartan)

Session 24

Lipid Lowering Drugs - Agents used in Hyperlipidemia

Text Assignment: Chapter 35

Learning Objectives:

1. Discuss metabolism of cholesterol and related lipoproteins, and their roles in atherosclerosis
2. Discuss management of hyperlipidemias/hyperlipoproteinemias.
3. Discuss mechanisms of action, adverse effects, potential interactions and specific clinical uses of lipid lowering drugs.

Drug Classes: Generic drug name (Brand name)

1. Inhibitors of Cholesterol Synthesis:
 - a. Lovastatin (Mevacor)
 - b. Pravastatin (Pravachol)
 - c. Simvastatin (Zocor)
 - d. Fluvastatin (Lescol)
 - e. Atorvastatin (Lipitor)
 - f. Rosuvastatin (Crestor)
2. Potentiators of Cholesterol Removal (alirocumab, evolocumab)
3. Nicotinic Acid (Niacin)
4. Bile Acid Sequestrants:
 - a. Cholestyramine (Questran)
 - b. Colestipol (Colestid)
5. Ezetimibe (Zetia)
6. The Fibrates:
 - a. Clofibrate (Atromid-S)
 - b. Gemfibrozil (Lopid)
 - c. Fenofibrate (Tricor)
7. Combination drugs:
 - a. Niacin/Simvastatin (Simcor)
 - b. Ezetimibe/Simvastatin (Vytorin)
 - c. Ezetimibe/Atorvastatin (Liptruzet)
 - d. Amlodipine besylate/Atorvastatin (Caudet)
8. Others:
 - a. Probucol (Lorelcol)
 - b. Orlistat (Xenical)

Session 25

Pharmacology of Ischemic Heart Disease: Vasodilator and Anti-Anginal Drugs Text Assignment: Chapter12

Learning Objectives:

1. Distinguish between classic angina (angina of effort), variant angina (angiospastic or Prinzmetal) and unstable angina.
2. List the major determinants of coronary blood flow, myocardial oxygen demand and supply.
3. Contrast the therapeutic and adverse effects of nitrates, beta-blockers and calcium channel blockers when used for angina.
4. Explain why the combination of a nitrate with a beta-blocker or a calcium channel blocker may be more effective than either alone.
5. Explain why the combination of a nitrate and sildenafil is potentially dangerous.
6. Contrast the effects of medical therapy and surgical therapy of angina.
7. Describe the significance of a first-pass effect for orally administered anti-anginal drugs.
8. Newer antianginal drugs: 1) pPox inhibitors (trimetazidine); 2) Ranolazine; 3) Ivabradine

Representative drugs:

NITRATES

1. Amyl nitrite
2. Nitroglycerin
3. Isosorbide dinitrate

BLOCKERS of Ca⁺⁺ CHANNELS

1. Diltiazem
2. Nifedipine
3. Verapamil

BLOCKERS of β - RECEPTORS

1. Propranolol
2. Timolol
3. Nadolol

OTHERS

1. Dipyridamole
2. Aspirin
3. Heparin
4. Glycoprotein IIb/IIIa

Session 26

Drug Treatment for Congestive Heart Failure Text Assignment: Chapter 13

Learning Objectives:

1. Describe the strategies and list the major drug groups used in the treatment of chronic heart failure.
2. Explain the effects and mechanisms of action of cardiac glycosides relative to mechanical and electrical properties of heart tissue.
3. List some positive inotropic drugs that have been investigated as digitalis substitutes.
4. Identify the major drug interactions of concern in patients taking cardiac glycosides.
5. Outline proposed steps in the management of cardiac glycoside toxicity.
6. Outline and explain the therapeutic steps in the treatment of acute and chronic heart failure.
7. Describe the other agents used frequently in the treatment of congestive heart failure: diuretics, vasodilators, ACE inhibitors, and other drugs that lack positive inotropic effects in heart failure.

Representative Drugs:

- | | |
|---|--|
| 1. Inotropic Therapy:
A. Cardiac glycosides (digoxin; digitoxin)
B. Sympathomimetics (dopamine, dobutamine)
C. Phosphodiesterase inhibitors
(amrinone, milrinone)
(hydralazine & isosorbide dinitrate) | 2. Arteriolar dilators/reduction of afterload
A. Direct vasodilators:
nitroprusside, minoxidil
hydralazine, diazoxide, Bidil |
| 3. <u>Venous dilators/relief of excessive preload:</u>
Diuretics (furosemide, hydrochlorothiazide, spironolactone, eplerenone, triamterene) prazosin, captopril, nitrates, nitroprusside | B. Angiotensin II System Antagonists:
1. ACE inhibitors: captopril, Chlorthalidone, enalapril
2. Angiotensin II receptor antagonists: losartan |
| 4. <u>Beta -adrenoceptor blockers:</u>
bucindolol, metoprolol, carvedilol, bisoprolol | C. Calcium channel blockers:
verapamil, diltiazem, nifedipine, felodipine, amlodipine |

Session 27

Antiarrhythmic Drugs Text Assignment: Chapter 14

Learning Objectives:

1. List common arrhythmias according to their origin.
2. Explain when arrhythmias should be treated?
3. Outline the basic mechanisms of arrhythmias – disturbances of impulse formation and conduction.
4. Describe the slow (calcium-dependent) and fast (sodium-dependent) responses and their alteration by antiarrhythmic drugs.
5. Classify antiarrhythmic agents based on their mechanisms of action.
6. Describe the major pharmacokinetic properties, therapeutic indications and adverse effects of the antiarrhythmic drugs listed below (see drug list).
7. Describe the beneficial and adverse interactions between antiarrhythmic drugs, and between antiarrhythmic drugs, and cardiac glycosides.

Representative Drugs:

CLASS 1A (blockers of fast Na⁺ channels)

- Procainamide • Quinidine • Disopyramide

CLASS 1B (weak block)

- Lidocaine • Mexiletine

CLASS 1C (pronounced block)

- Flecainide • Propafenone

CLASS 2 (β-Blockers)

- Propranolol • Esmolol • Metoprolol

CLASS 3 (blockers of K⁺ channels)

- Amiodarone • Dofetilide • Bretylium • Sotalol • Ibutilide • Dronedarone

CLASS 4 (blockers of Ca²⁺ channel)

- Verapamil • Diltiazem

OTHER DRUGS

- Adenosine • Digoxin • Magnesium • Potassium

Session 28

Renal Pharmacology - Diuretics & Other Drugs Affecting Renal Function Text Assignment: Chapter 15

Teaching Goals:

Present information to impart to the student an understanding of:

1. The actions (both systemic and renal) of diuretic drugs relative to basic functions of the nephron.
2. Concepts and practicalities of the clinical use diuretics.

Learning Objectives:

1. Identify the site, mechanism, and consequences of action of diuretic drugs.
2. Outline the effects of diuretics on urine volume, urine pH and urine ion composition.
3. Explain physiologic compensation mechanisms following decreased plasma volume.
4. Describe the relationship of diuretic effects and alkalosis.
5. Identify the principal adverse effects of diuretic drugs.
6. Apply knowledge of diuretic drugs to clinical questions and case scenarios.

Representative Drugs:

Acetazolamide
Amiloride
Demeclocycline
Desmopressin
Furosemide
Hydrochlorothiazide
Mannitol
Probenecid
Spironolactone
Triamterene

SEMESTER 2

Session 29

Respiratory Drugs - Agents Used in Asthma, COPD Text Assignment: Chapter 20

Learning Objectives:

1. Discuss the strategies of drug treatment of asthma
2. Discuss the main therapeutic interventions available for treating asthma. a) Identify therapies for long-term control of inflammatory reactions. b) Identify therapies for quick relief of bronchoconstriction.
3. Discuss drugs used in treatment of COPD
4. Identify drugs used to treat symptoms of cough and congestion
5. Discuss mechanisms of action, adverse effects, and specific clinical uses of the prototype asthma drugs
6. Apply knowledge of respiratory drugs to clinical questions and case scenarios.

Representative Drugs: Asthma -

long-term control

Mast cell stabilizers - Cromolyn [Intal]

Nedocromil [Tilade]

Corticosteroids - Beclomethasone [Beclovent]

Prednisone

Leukotriene antagonists - Zafirlukast [Accolate]
Zileuton [Zyflo] Montelukast [Singulair]

Asthma - quick-relief

β_2 -adrenergic agonists - Albuterol [Proventil]
Pirbuterol [Maxair]

Terbutaline [Brethine]
*Salmeterol [Serevent]

Muscarinic antagonists - Ipratropium [Atrovent]

Methylxanthines - Theophylline

Aminophylline

Sessions 30 & 31

Toxicology 1 & 2

Text Assignment: Chapters 56, 57, 58

Learning Objectives:

1. Explain the differences between systemic and local toxic effects.
2. Contrast the basic principles in toxicokinetics and pharmacokinetics.
3. Give examples of organ-specific toxicities and explain how these specificities are related to physiological or biochemical properties of those organs.
4. Discuss the role of biotransformation and detoxification mechanisms in determining the chemical's toxicity.
5. Give examples of mechanism-based treatments for different types of toxins.
6. Explain the toxic effects of organophosphates, carbon monoxide, cyanide, methanol, lead, mercury, and iron. Specify symptoms associated with each of these toxins. Discuss physiological defense mechanisms and treatments to diminish their toxicities.
7. Discuss important pharmacokinetic concerns and limitations of effectiveness in therapeutic use of chelating agents.
8. Discuss the role of pharmacogenetics in the variation in toxicity between individuals exposed to the same chemicals.
9. Describe general therapeutic procedures for management of chemical toxicity.
10. Describe principal concerns in supportive treatment of poisoned patients, regardless of toxin.

Session 32

Antipsychotic Drugs Text Assignment: Chapter 29

Learning Objectives:

1. Explain why antipsychotic drugs are called "Neuroleptics."
2. Classify the phenothiazines and compare their pharmacologic potencies.
3. Explain the dopamine hypothesis of Schizophrenia.
4. Describe the neuroendocrine effects of the antipsychotic drugs.
5. In addition to the treatment of schizophrenia list other therapeutic indications for antipsychotic drug therapy.
6. Describe the pharmacologic interactions of antipsychotic agents with guanethidine and tricyclic antidepressants.
7. What is tardive dyskinesia? What is the cause of this condition? Can it be treated?
8. Each of the three classes of drugs listed below is used with similar efficacy in the treatment of schizophrenia. They are also used to treat nausea and vomiting as well as amphetamine- and LSD- induced psychosis. Compare incidence of side effects produced by each, paying particular attention to sedation, hypotension and extrapyramidal symptoms.
 - A. Phenothiazines; Chlorpromazine, thioridazine, trifluoperazine
 - B. Thioxanthenes: Chlorprothixene
 - C. Butyrophenones: haloperidol
9. Statement on Clozapine and related drugs.

Representative Drugs:

Other drugs will be introduced during class.

Session 33

Antidepressants and Lithium Text Assignment: Chapters 29& 30

Learning Objectives:

1. List the therapeutic indications for use of antidepressants.
2. List the 3 groups of antidepressant drugs and indicate their mechanisms of action.
3. List any specificity the tricyclics may exhibit toward a particular neurotransmitter.
4. How is the antidepressant drug amoxapine related to the antipsychotic drugs?
5. Describe the reasons for the latency of onset of effects of antidepressants.
6. What are the consequences of the effects of tricyclic antidepressants on the Autonomic Nervous System and cardiovascular system?
7. What are the potential interactions of the tricyclic antidepressants with guanethidine, phenylephrine and MAO inhibitors?
8. List the CNS effects of lithium. Why should the plasma levels of lithium be monitored during the course of treatment?
9. What is the relationship between lithium and thyroid function?

Representative Drugs:

Tricyclic Antidepressants

Amitriptyline
Sedipramine
Imipramine
Nortriptyline

Second Generation Antidepressants

Amoxapine
Bupropion
Maprotiline
Trazadone

SSRIs

Fluoxetine
Paroxetine
Sertraline

Third Generation Antidepressants

Fluvoxamine
Nefazodone
Venlafaxine

MAO Inhibitors

Isocarboxazid
Phenelzine
Tranylcypromine

Mood Stabilizers

Carbamazepine
Lithium Carbonate
Valproic Acid

Session 34

Sedative/Hypnotics Drugs Used to Treat Anxiety & Insomnia Text Assignment: Chapters 22

Learning Objectives:

1. Define sedation, hypnosis and anxiety.
2. Identify the active ingredients in several over-the-counter hypnotic products used by individuals to aid in the induction of sleep at bedtime. To what classes of drugs do these active ingredients belong?
3. What are the mechanisms of action of the barbiturate sedative-hypnotics? The benzodiazepines?
4. Name the only sedative-hypnotic agent that does not exert additive central nervous system depression when administered with other drugs.
5. Name the sedative-hypnotic agent that is not metabolized by hepatic microsomal enzymes but can stimulate the enzymes to increase the rate of metabolism of other drugs.
6. Barbiturates can be classified according to their duration of action. List the barbiturates according to this classification.
7. Describe the effects of cimetidine on the biotransformation of certain benzodiazepines.
8. Explain the relationship between the effect of hypnotics on REM sleep and hangover.
9. All barbiturates have anticonvulsant effects. Explain why only three of these barbiturates are used as anticonvulsants. List the three drugs.
10. Explain why barbiturates are contraindicated in acute intermittent porphyria.
11. Explain why the administration of barbiturates can lead to vitamin D deficiency in some children.

Representative Drugs:

Benzodiazepines:

Alprazolam
Clorazepate
Chlordiazepoxide
Clonazepam
Demoxepam
Diazepam
Flurazepam
Lorazepam
Midazolam
Temazepam
Triazolam

Barbiturates

Pentobarbital
Phenobarbital

Secobarbital

Thiopental

Non-Barbiturates

Chloral hydrate

Paraldehyde

Anxiolytic Agent

Buspirone

Selective Hypnotics

Zolpidem
Zaleplon
Zopiclone

Benzodiazepine Antagonist

Flumazenil

Session 35

Drugs used in the treatment of disorders of coagulation.

Drugs used in the treatment of anemias.

Text Assignment: Chapters 33 & 34

Learning Objectives: Coagulation Disorders:

1. Review basic physiologic steps and interactions involved in platelet activation and aggregation and in blood coagulation/fibrinolysis.
2. Discuss mechanisms of action of antithrombic drugs, anticoagulant drugs, fibrinolytic drugs and agents used in treating bleeding disorders.

Representative drugs used for the treatment of coagulation and bleeding disorders will be presented as part of the class.

Representative Drugs:

Coagulation Disorders:

1. Heparin (Calciparine) and its derivatives
2. Direct thrombin inhibitors
3. Active Factor X (Xa) Inhibitors
4. Plasminogen Activators:
 - Tissue plasminogen activator (tPA)
 - Alteplase
 - Reteplase
 - Tenecteplase
 - Streptokinase
 - Anistreplase (complex of streptokinase & plasminogen)
 - Urokinase
5. Warfarin (Coumadin)
6. GP (Glycoprotein) IIb/IIIa inhibitors
7. Adenosine diphosphate (ADP) inhibitors
8. PDE₃ inhibitors
9. Aspirin and other NSAIDS
10. Protease activated receptor-1 (PAR-1) inhibitor: Vorapaxar

Bleeding Disorders:

1. Vitamin K
2. Protamine sulfate
3. Aminocaproic acid
4. Tranexamic acid
5. Idarucizumab (Praxbind)

Learning Objectives: Anemias

1. Discuss the absorption, distribution, and storage of iron.
2. Discuss iron deficiency, folic acid deficiency, and vitamin B₁₂ deficiency.
3. Discuss how megaloblastic anemia of vitamin B₁₂ deficiency can be partially corrected with folic acid.
4. Discuss the importance of distinguishing vitamin B₁₂ deficiency from folic acid deficiency, and how neurologic damage may occur if vitamin B₁₂ deficiency is incorrectly treated with folic acid.
5. Discuss indications for the use of erythropoietin and other growth factors.

Representative Drugs:

1. Iron
2. Deferoxamine
3. Vitamin B₁₂
4. Folic acid

5. Hematopoietic Growth Factors

Erythropoietins

Epoetin alpha

Darbepoetin - modified form of erythropoietin with longer half-life

Myeloid Growth Factors

Filgrastim

Pegfilgrastim – conjugate of filgrastim to increase half-life

Sargramostim

Megakaryocyte Growth Factors

Interleukin-11 (Oprelvekin)

Thrombopoietin - (Analogues - Romiplostim; Eltrombopag)

Session 36

Gastrointestinal Agents Text Assignment: Chapter 62

Learning Objectives:

1. Discuss the physiological control mechanisms in the gastrointestinal tract that can be altered by pharmacological agents.
2. Discuss how emesis is affected by such a wide variety of pharmacological agents acting at various receptor types.
3. Discriminate between drugs acting at the chemo-receptor trigger zone and those that are effective in the treatment of motion sickness.
4. Describe the role of antibiotic therapy in treating peptic ulcers.
5. Identify the physiological controls of gastric acid secretion and describe how therapeutic agents can alter these controls.
 - a. Describe the mechanism of action in decreasing acid secretions of proton-pump inhibitors and why these are more efficacious than other treatments.
 - b. Describe the role of histamine in gastric acid secretion and explain the effectiveness of H₂-receptor antagonists in blocking secretions.
 - c. Discuss the factors that are important in comparing the efficacy of various antacids.
6. Describe the mechanisms of action of prokinetic agents in terms of the receptors involved.
7. Site examples of anti-ulcer treatments that should not be combined. Site examples of combined treatments that are advantageous. Explain these in terms of drug interactions:
 - a. H₂-antagonists
 - b. antacids
 - c. barrier agents
8. Explain the actions of opiates in acute treatment of diarrhea.
9. Describe the principal contraindications for laxatives that are also of concern with antacids.
10. Discuss the difference in therapeutic goals for Intestinal Bowel Syndrome (IBS) and Irritable Bowel Disease (IBD) and how these are achieved.

Representative Drugs

Anti-emetics

Scopolamine [Transderm]
Dimenhydrinate [Dramamine]
Meclizine [Bonine]

Promethazine [Phenergan]
Diphenhydramine [Benadryl]

Dronabinol [Marinol]
Metoclopramide [Reglan]

Ondansetron [Zofran]

Palonosetron [Aloxi]

Granisetron [Kytril]

Aprepitant [Emend]

Emetics

Apomorphine

Anti-ulcer

Omeprazole [Prilosec]
Rabeprazole [Aciphex]
Cimetidine [Tagamet]
Famotidine [Pepcid]

Lansoprazole [Prevacid]
Pantoprazole [Protonix]
Ranitidine [Zantac]
Nizatidine [Axiid]

Esomeprazole [Nexium]

aluminum hydroxide [Amphojel]
sodium bicarbonate [Alka-Seltzer]

magnesium hydroxide [Milk of Magnesia]
calcium carbonate [Titalac]

Sucralfate [Carafate]

Bismuth subsalicylate [Peptobismol]

Misoprostol [Cytotec]

Metoclopramide [Reglan]

Erythromycin

Anti-diarrheals

Loperamide [Imodium]
Kaolin, pectin [Kaopectate]

Diphenoxylate [Lomotil]
Bismuth subsalicylate [Peptobismol]

Laxatives

psyllium [Metamucil]
bisacodyl [Dulcolax]
cascara

lactulose [Cephulac]
danthron [Modane]
castor oil

IBS treatments

alosetron [Lotonex]
tegaserod [Zelnorm]

lubiprostone [Amitiza]

IBD treatments

5-aminosalicylic acid
budesonide [Entocort]
azathiopurine [Imuran]
methotrexate [Rheumatrex, Trexall]

infliximab [Remicade]
adalimumab [Humira]
certolizumab [Cimzia]

Sessions 37 & 38

Drugs of Abuse 1 and 2 Text Assignment: Chapter 23 & 32

Learning Objectives:

1. Describe what is meant by drug dependence and drug addiction.
2. Describe what is meant by drug tolerance, including innate, acquired, and conditioned.
3. Identify the anatomical locations and pharmacological receptors involved in drug addiction, including euphoria and reinforcement.
4. Identify the three major receptor-based mechanisms by which drugs of addiction enhance dopamine levels.
 - a. Indicate which of these mechanisms are involved in the addictive effects of:
 - b. Discuss the pharmacological basis for toxicity of the addictive drugs:
 - c. Discuss available pharmacological treatments for addiction to:
 - 1) Opioids
 - 2) Cannabinoids
 - 3) Nicotine
 - 4) Ethanol
 - 5) Cocaine
 - 6) Amphetamine
5. Describe the enzymatic metabolism of ethanol.
6. Discuss the genetic bases for susceptibility to alcohol addiction and how these are related to treatments for addiction.

Session 39

Endocrine Pharmacology I

A: Pharmacology of Hypothalamic and Pituitary Hormones

Text Assignment: Chapters 37, 38, 42

General Principles of Hormone Action

Learning Objectives:

1. Define three general goals of endocrine therapy; list specific disorders and examples of pharmacologic agents used to achieve these goals.
2. List the principal types of hormones based upon structure of the hormone and location of receptors.
3. Classify major hormones according to their signal transduction pathways.

Hypothalamus/Pituitary

1. List hormones and/or neurotransmitter of hypothalamic origin that control hormone secretion from the anterior pituitary.
2. Identify anterior pituitary hormones and be able to describe: a) their general structure; b) regulation and pattern of release (including alteration by drugs); c) target organ; d) target organ hormone(s) actions; e) medical problems related to hypo- and hyper-secretion; f) clinical uses; and g) side effects when used as therapeutic agents.
3. List posterior pituitary hormones and be able to outline: a) their general structure; b) regulation and pattern of release (including alteration by drugs); c) target organ; d) target organ hormone(s) actions; e) medical problems related to hypo- and hyper-secretion; f) clinical uses; and g) side effects when used as therapeutic agents.

Representative Drugs: Hypothalamus/Pituitary

1. gonadotropin releasing hormone (Gonadorelin, Lutrepulse)/goserelin (Zoladex)
2. leuprolide (Lupron)/nafrelin (Synarel)
3. gonadotropins, (LH, FSH, hCG)/menotropins (hMG, Pergonal)/urofollotropin (Metrodin)
4. growth hormone (somatropin, Humatrope7/Nutropin AQ)
5. octreotide (Sandostatin)
6. prolactin
7. bromocriptine (Parlodel)/cabergoline (Dostinex)
8. corticotropin (ACTH, Cosyntropin)
9. oxytocin (Pitocin)
10. vasopressin (ADH)/desmopressin (DDAVP)
11. tolvaptam (Samsca)

Session 39 Continued

B: Pharmacology of Thyroid and Parathyroid Hormones

Thyroid/Parathyroid

Learning Objectives:

1. Describe the synthesis of T₄ (and T₃) and relate the sites and the mechanism(s) of action of: levothyroxine, propylthiouracil (PTU), and potassium iodine, ¹³¹I.
2. Compare the pharmacokinetics of levothyroxine (T₄) and triiodothyronine (T₃); explain the rationale for utilizing T₄ over T₃ in treating hypothyroidism.
3. List the rationale(s) for the following therapeutic indications: beta-antagonists for hyperthyroidism; value of TSH levels as an index of replacement therapy.
4. Describe the adverse effects and the precautions in use of the thioamides (PTU & methimazole)

Representative Drugs: Thyroid/Antithyroid Agents

- | | |
|--|--------------------------------|
| 1. levothyroxine sodium (Synthroid, T ₄) | 5. methimazole (Tapazole) |
| 2. liothyronine (Cytomel, T ₃) | 6. propranolol (Inderal) |
| 3. potassium iodide (SSKI) | 7. I ¹³¹ (Iodotope) |
| 4. proplthiouracil (PTU) | |

Parathyroid/Ca²⁺ (Agents that affect bone and mineral homeostasis)

1. Describe the role(s) of PTH, calcitonin and vitamin D effects on bone, gut and kidney with respect to Ca²⁺ homeostasis.
2. Provide the therapeutic rationale for: a) Ca²⁺ salts and vitamin D in hypocalcemia and b) calcitonin, bis-phosponates, furosemide for hypercalcemia; c) Vitamin D₃ in renal failure
3. List the adverse effects of: vitamin D deficiency and excess; bisphosphonates.

Representative Drugs: Parathyroid/Ca²⁺ Agents

1. calcitriol (1, 25-dihydroxy-vitamin D)
2. calcitonin
3. parathyroid hormone (PTH, Teraparitide)
4. alendronate (Fosamax) pamidronate, and zoledronic acid (bis-phosphonates)
5. prednisone
6. calcium gluconate/carbonate
7. sodium fluoride
8. furosemide (Lasix)
9. denosumab (Prolia)

Session 40

Endocrine Pharmacology II A: Pharmacology of The Adrenal Glands/Corticosteroids Text Assignment: Chapters 39 and 40

Corticosteroids (Glucocorticoids/Mineralocorticoids)

Learning Objectives:

1. Describe the major steps of biosynthesis and the structure-activity relationship of endogenous and synthetic corticosteroids.
2. Describe the feedback regulation of corticosteroid (glucocorticoids and mineralocorticoids) synthesis and release and related this to rationale for alternate day therapy.
3. Describe significance of corticosteroid disposition (protein binding, elimination, significance of enzyme induction, situations requiring dosage adjustments) as related to their use as therapeutic agents.
4. Outline the cellular mechanism(s) of action of corticosteroids.
5. Describe the pharmacologic (therapeutic) and adverse effects of corticosteroids (glucocorticoids/mineralocorticoids).
6. List the therapeutic uses and explain rationale for the uses of glucocorticoids and mineralocorticoids in various diseases.
7. List contraindications to glucocorticoid/mineralocorticoid use.
8. Explain the concepts demonstrated by drugs used diagnostically for assessing and treating hypothalamic pituitary adrenocortical dysfunction e.g., metyrapone, dexamethasone, ACTH, ketoconazole, and spironolactone.

Representative Drugs: Oral/Parenteral Glucocorticoids

1. hydrocortisone (cortisol; Hydrocortone)
2. prednisone (Deltasone)
3. dexamethasone (Decadron)

Aerosol Glucocorticoids

1. beclomethasone (Celestone/Vancenase AQ (Nasal Spray)
2. triamcinolone (Azmacort)

Mineralocorticoids

1. fludrocortisone (Florinef)
 2. aldosterone
- Aldosterone Antagonist
1. spironolactone (Aldactone)

Adrenal Cortex Inhibitors

1. metyrapone (Metopirine)
2. aminoglutethimide (Cytadren)
3. ketoconazole (Nizoral)
4. mitotane (Lysodren)

Session 40 Continued:

Endocrine Pharmacology II B: Androgens and Drugs Affecting Male Sexual Function

Learning Objectives:

1. List the biological actions of androgens (anabolic steroids).
2. List the therapeutic uses and abuses of androgenic/anabolic steroids.
3. Describe the side effects of androgens (anabolic steroids).
4. List the classes of drugs used to treat benign prostatic hyperplasia (BPH) based upon their mechanism(s) of action.
5. State the side effects of drugs used to treat BPH.
6. Describe the therapeutic approaches to erectile dysfunction and the contraindications for therapeutic agents.

Representative Androgens and related drugs

1. testosterone
2. oxandrolone (Anavar)
3. finasteride (Proscar)
5. spironolactone (Aldactone)
6. danazol (Danocrine)
7. flutamide (Eulexin)

BPH Agents

α_1 Adrenoceptor Antagonists

1. terazosin (Hytrin)
2. doxazosin (Cardua)
3. tamsulosin (Flomax)

5 α -Reductase Inhibitor

1. finasteride (Proscar)

Erectile Dysfunction

1. alprostadil (Caverject)
2. sildenafil (Viagra)

Session 41

Endocrine Pharmacology III

A: Pharmacology of Estrogens, Progestins, and Drugs Affecting Peripartum Management Text Assignment: Chapter 40

Learning Objectives:

1. List the biologic actions and adverse effects of estrogens and progestins and describe hormonal and other pharmacologic approaches to fertility control/contraception
2. Describe the therapeutic uses of estrogen and progestin derivatives.
3. Outline the absolute and relative contraindications that preclude the use of oral contraceptives.
4. Describe the actions of oxytocin, ergot derivatives and prostaglandins on uterine function.
5. Identify drugs that may be used to suppress uterine contractions and describe their mechanism(s) of action.
6. Identify various drugs that can be used to induce a therapeutic abortion.

Representative Drugs: Estrogens/progestins and related drugs

1. norethindrone (Norlutin), levonorgestrel (Norplant), medroxyprogesterone (Depo-Provera), drospirinone (Yasmin, others)
2. ethinyl estradiol (Estinyl), mestranol and oral contraceptive packs
3. clomiphene
4. tamoxifen (Nolvadex)/anastrozole (Arimidex)
5. diethylstilbestrol [DES]
6. mifepristone (RU486)

Uterine stimulants

1. oxytocin (Pitocin)
2. ergonovine (Ergotrate)
3. PGF₂ α , dinoprost (Prostin) PGE₂,

Uterine relaxants

1. MgSO₄
2. ethanol

Session 42

Endocrine Pharmacology IV

Pharmacology of The Pancreatic Hormones & Antidiabetic Drugs and Endocrine Review

Text Assignment: Chapter 41

Learning Objectives: Hormones and Drugs

1. Outline the predominant effects of insulin, glucagon, growth hormone, corticosteroids, epinephrine on intermediary metabolism.
2. Describe the differences between type I and type II diabetes relative to recommended pharmacologic therapies.
3. Describe the pharmacokinetics and pharmacodynamics of oral sulfonylureas, metformin, and thiazolidinediones (pioglitazone) and SG
4. List uses and adverse effects of sulfonylureas.
5. Describe different types of insulin preparations (including lispro and glargine insulin) and the pharmacokinetics (onset, duration, and peak of action, clearance).
6. Contrast the action(s) and use(s) of incretin analogs, dipeptidyl peptidase inhibitors, and
- 7.
8. Know the side effects of insulin therapy; identify agents used to treat hypoglycemic episodes.
9. Describe the potentially harmful effects of ethanol and β -receptor antagonists in a diabetic patient and provide explanations for the adverse effects produced by interactions with insulin and/or oral sulfonylureas and biguanides with other drugs.

Oral Agents (Type II Diabetes: NIDDM)

1. glipizide (Glucotrol) (sulfonylurea)
2. metformin (Glucophage)
3. pioglitazone (Actos)
4. canagliflozin (Invokana)(SGLT2 inhibitor)
5. acarbose (Precose)
6. exenatide (Byetta) (incretin mimetic),
semaglutide (Ozempic)
7. sitagliptin (Januvia) (DPP4 inhibitor)
8. tirzepatide (Mounjaro) GLP-1 and GIP inhibitor

Insulins (Type I/Type II Diabetes; IDDM)

1. rapid-acting(Lispro insulin [Humalog])
2. short-acting (regular, Humulin)
3. intermediate-acting (Lente, NPH)
4. long-acting (glargine insulin/Lantus)

Hyperglycemic Agents

1. glucagon
2. diazoxide (Proglycem)

Session 43

Drugs Used in Treatment of Gout/Arthritis [Disease-Modifying Antirheumatic Drugs (DMARDs)]

Text Assignment: Chapter 36

Learning Objectives:

1. Describe the agents used for rheumatoid arthritis and their mechanisms of actions.
2. Discuss the toxicities and adverse effects of agents used for rheumatoid arthritis.
3. Identify the primary drugs used for gout and their mechanisms of action,
4. Discuss the adverse effects and contraindications to their usage.

Representative Drugs:

Disease-Modifying Antirheumatic Drugs:

1. anakinra
2. etanercept
3. hydroxychloroquine
4. infliximab

Drugs used in the treatment of gout:

1. allopurinol
2. colchicine
3. probenecid
4. sulfinpyrazone

Session 44

Local Anesthetics Text Assignment: Chapter 26

Learning Objectives:

1. What are the core chemical structural entities of most local anesthetics?
2. Most local anesthetics fall into two chemical classifications:

What is the basis for these classifications? Why would one class be preferred over the other?
3. What is the mechanism of action of all local anesthetics? What happens to the nerve action potential and resting membrane potential following application of a local anesthetic?
4. How does the pH of the surrounding medium alter the effectiveness of a particular local anesthetic?
5. What is the relationship between the fibre type and loss of function attributed to the actions of local anesthetics according to the sensitivity of block?
6. What is the general effect of most local anesthetics upon the vasculature? How can the duration of action of a local anesthetic be modified?
7. What are the potential toxic effects of local anesthetics?
8. Which local anesthetic should you choose for epidural anesthesia?

Representative Drugs:

Esters

Benzocaine
Cocaine
Procaine
Tetracaine

Amides

Bupivacaine
Etidocaine
Lidocaine
Mepivacaine
Prilocaine

Sessions 45 and 46

Inhalation & Intravenous Anesthetics Text Assignment: Chapter 25

Learning Objectives

1. What are the purposes of general anesthesia?
2. Define the term: "balanced anesthesia"
3. List the general anesthetics that sensitize the heart to catecholamine induced arrhythmias.
4. What is the mechanism of action of both inhalation and intravenous anesthetic agents?
5. Define blood-gas partition coefficient. Compare N₂O, and halothane.
6. What is the relationship between blood solubility and a) speed of onset of anesthesia b) potency of an anesthetic agent?
7. What is malignant hyperthermia? What is the mechanism of action of the drug used to treat this condition?
8. What are the disadvantages of a slow-onset anesthetic? What can be done to overcome these disadvantages?
9. Explain the cause and the result of the second gas effect and diffusion hypoxia.
10. Why does the concentration of an anesthetic in the brain often rise more rapidly than in the rest of the body?
11. List 4 drugs used to produce intravenous anesthesia.
12. Explain the term "dissociative anesthesia". What drug produces this type of anesthesia?
13. What drug combination produces the so-called neuroleptic anesthesia?
14. Outline the situations that require the use of other agents before or during anesthesia. What are the particular actions for which each preanesthetic agent is used?
15. List the general anesthetics that increase sympathetic nervous system activity.

Representative Drugs:

General (Inhalation) and Intravenous Anesthetic Agents, Premedicants and Adjuvants

Inhalation Anesthetics

Diethyl Ether
Enflurane
Halothane
Isoflurane
Methoxyflurane
Nitrous Oxide
Sevoflurane
Desflurand

Intravenous Anesthetics

Diazepam
Properidol
Etomidate
Fentanyl
Ketamine
Midazolam
Propofol (Diprivan)
Thiopental

Premedicants & Adjuvants

Sedatives

Diazepam
Midazolam
Pentobarbital
Secobarbital

Anticholinergics

Atropine
Glycopyrrolate
Scopolamine

Antihistamines

hydroxyzine
Promethazine

Opioid (Narcotic) Analgesics

Fentanyl
Meperidine

Butyrophenones

Droperidol
Haloperidol

Muscle Relaxants

Decamethonium
Succinylcholine
Atracurium
d-Tubocurarine
Pancuronium
Vecuronium

Session 47

Opioid Analgesics & Antagonists Text Assignment: Chapter 31

Read chapter 31 in the text to supplement your notes. Focus drugs in this category will be discussed in class. Read clinically relevant article(s).

The following teaching and learning goals and objectives serve also as a lecture outline.

Teaching Goals:

Present information to impart to the student an understanding of;

1. The pharmacokinetics and pharmacodynamics of opioid agonists, discussing morphine as the illustrative standard.
2. Concepts and practicalities of the clinical use of opioid agonists and antagonists.

Learning objectives: Following this presentation, the student should be able to:

1. Explain how the physico-chemical properties of morphine contribute and relate to its pharmacokinetic parameters.
2. Illustrate (stylistically) neural pathways involved in the sensation of pain and actions of opioid analgesics and other drugs likely to promote analgesic actions along these pathways.
3. Identify key opioid receptor types and pharmacodynamic actions attributed to the receptor types.
4. Classify opioid agonists according to receptor specificity.
5. Cite major pharmacodynamic actions of morphine relative to its action(s) regarding the CNS, CV system, GI & GU tracts, and neuro-endocrine effects.
6. Differentiate pharmacokinetic and pharmacodynamic differences between and among select opioid agents.
7. Differentiate pharmacokinetic and pharmacodynamic differences between and among select opioid agents.
8. Explains the difference(s) between/among opioid antagonists and their respective use(s).
9. Discuss current issues and trend in the use of opioid agonists in the management of chronic pain. Representative drugs will be presented as part of class discussion.

Session 48

Drugs Used to Treat Migraines Text Assignment: Chapter 16

Learning Objectives:

1. Discuss the proposed mechanisms of action of the various migraine treatments.
2. Outline the therapeutic goals of treatment for migraine.
 - a) Identify therapies for acute treatment of migraine.
 - b) Identify therapies for prophylactic treatment of migraine.
3. Describe the mechanism of action, indications, and toxicity of triptans.
4. Describe the major clinical applications and toxicities of the ergot drugs.

Representative Drugs:

5-HT	Ondansetron [Zofran]	Cyproheptadine [Periactin]
antagonists -	Ergotamine [Ergomar][Cafergot] [Sansert]	Dihydroergotamine [DHE] Methysergide
5-HT	Sumatriptan [Imitrex]	
agonists -	Cisapride [Propulsid]	
β-adrenergic	Propranolol [Inderal]	Atenolol [Tenormin]
antagonists -	Timolol [Blocadren]	Metoprolol [Lopressor] Nadolol [Corgard]
Tricyclic	Amitriptyline [Elavil]	
antidepressants -	Nortriptyline [Pamelor]	
Monoamine oxidase		
inhibitors -	Phenelzine [Nardil]	Isocarboxazid [Marplan]

Session 49

Anti-Parkinsonism Drugs and Agents Used in Neurodegenerative Disorders Text Assignment: Chapter 28

Learning Objectives:

1. What is the neurochemical defect in Parkinsonism and how does bromocriptine and L-dopa alleviate the defect?
2. List the advantages of the use of carbidopa in the treatment of the disease.
3. Why should the use of pyridoxine-containing medication be avoided during the treatment of Parkinsonism with L-dopa.
4. What are the mechanisms of action of deprenyl? Amantadine?
5. Explain the mechanism(s) of action and adverse effects associated with bromocryptine.

Representative Drugs:

Drugs used in treating Parkinson's disease:

Dopaminergics

Amatadine
Apomorphine
Bromocriptine
Carbidopa
Deprenyl (Selegiline)
Levodopa
Pergolide

Central Anticholinergics

Benztropine
Diphenhydramine
Trihexyphenidyl

Contraindicated

Pyridoxine (vitamin B6)

Drugs used in treating other movement disorders:

Propranolol
Trientine
Penicillamine

Session 50

Anti-seizure (Anti-epilepsy) Drugs

Text Assignment: Chapter 24

Learning Objectives:

1. Differentiate between the multitude of seizure disorders
2. Understand that different seizure disorders require different pharmacological therapies
3. Briefly describe the etiology of epilepsy
4. Understand that both increased excitation and diminished inhibition disrupt the normal balance between excitation and inhibition, and that different drugs target excitatory or inhibitory mechanisms
5. Recognize the mechanisms of action, pharmacokinetic properties, and major side effects of anti-seizure medications
6. Recognize the potential drug interactions of anti-seizure medications

Anti-seizure medications for epilepsy, status epilepticus, Absence (petit mal) seizures, and infantile spasms:

Acetazolamide

ACTH

Cannabidiol

Carbamazepine

Cenobamate

Clobazam

Clonazepam

Diazepam

Ethosuximide

Everolimus

Felbamate

Fenfluramine

Gabapentin

Lacosamide

Lamotrigine

Levetiracetam

Lorezepam

Midazolam

Perampanel

Phenobarbital

Phenytoin, fosphenytoin

Pregabalin

Primidone

Retigabine

Rufinamide

Stiripentol

Tiagabine

Topiramate

Trimethadione

Valproic acid

Vigabatrin

Zonisamide

Session 51

Drugs Used to Treat Ocular Disorders – Glaucoma & ARMD Text Assignment: Chapters 6, 7 & 10

Learning Objectives:

1. Know the basic pharmacology of the eye
2. Know some of the common ocular disorders
3. Know the pharmacokinetics of ocular therapeutic agents
4. Identify risk factors for the development of primary open-angle glaucoma (POAG) and acute angle-closure glaucoma.
5. Know the treatment goals for patients with primary open-angle glaucoma (POAG), and acute angle-closure glaucoma.
6. Know the mechanisms of action and adverse effects of the different categories of agents used to treat POAG.
7. Describe the types of age-related macular degeneration (ARMD)
8. Know the treatment options for wet ARMD

Representative Drugs Used for Treatment of POAG:

Prostaglandin analogues

- Latanoprost, bimatoprost, travoprost, tafluprost

Beta Blockers

- Timolol, betaxolol, carteolol, levobunolol, metipranolol

Alpha₂ adrenergic receptor agonists

- Apraclonidine, brimonidine

Cholinomimetics

- Pilocarpine, carbachol

Carbonic anhydrase inhibitors

- Dorzolamide, brinzolamide

Rho kinase inhibitors

- Netarsudil

Treatment of ARMD

VEGF-A inhibitors

Antioxidant combinations

Session 52

Chrono-Pharmacology

General Principles of Chronopharmacology

1. Define the general characteristics of circadian rhythms;
2. Define the role of circadian rhythm on organ functions;
3. List specific disorders that may benefit from a treatment that consider the time of the day for administration of drugs
4. List the advantages of considering the time of the day to the administration of different drugs

Representative Chronobiotic Drugs:

1. Melatonin
 2. Melatonin agonists (Talsimelteon/ramelteon)
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Mini Board
NBME Subject Area Exam
20% of Course Grade
MUCH SUCCESS

Medical Pharmacology and Toxicology

Class Meeting Times: 8:30 a.m. to 10:30 a.m. (unless otherwise indicated on the calendar)

Lecture Site: **Second Year Classroom**

Date	Session	Course Topic	Instructor
GENERAL PRINCIPLES / PHARMACOLOGY FOUNDATIONS			
Aug. 13	1	Gen. Principles I (Pharmacodynamics)	Dr. Russell Randall
Aug. 15	2	Gen. Principles II (Pharmacokinetics I)	Dr. Cropp /Dr. Shepherd
Aug. 20	3	Gen. Principles III (Pharmacokinetics II)	Dr. Cropp /Dr. Shepherd
Aug. 22	4	Gen. Principles IV (Pharmacokinetics III)	Dr. Cropp /Dr. Shepherd
Aug. 26	5	Intro. to Pharmacogenomics	Dr. Cropp
Sept. 03	Tuesday	EXAM I (10:30 a.m. - 12:30 p.m.)	
PHARMACOLOGY FOUNDATIONS (contd.) / IMMUNITY			
Sept. 05	6	Intro. to Autonomic Pharmacology	Dr. Shepherd
Sept. 06	7	Cholinergic Agonists	Dr. Shepherd
Sept. 09	8	Cholinergic Antagonists	Dr. Shepherd
Sept. 11	9	Adrenergic Agonists	Dr. Shepherd
Sept. 13	10	Adrenergic Antagonists	Dr. Shepherd
Sept. 16	11	Treatment of Inflammation I (Eicosanoids)	Dr. Eatman-Daniels
Sept. 18	12	Treatment of Inflammation II (NSAIDS)	Dr. Eatman-Daniels
Sept. 25	13	Autacoids (10:30 am - 12:30 pm)	Dr. Eatman-Daniels
Sept. 26	14	Immunopharmacology (10:30 am - 12:30 pm)	Dr. Cropp/Dr. Sloviter
Oct. 03	Thursday	EXAM II (10:30 a.m. - 12:30 p.m.)	
CHEMOTHERAPY			
Oct. 07	15	Cancer Chemotherapy I	Dr. Cropp /Dr. Sloviter
Oct. 10	16	Cancer Chemotherapy II	Dr. Cropp /Dr. Sloviter
Oct. 15	17	Anti-Infectives I (10:30 am - 12:30 pm)	Dr. Russell Randall
Oct. 16	18	Anti-Infectives II (10:30 am - 12:30 pm)	Dr. Russell Randall
Oct. 17	19	Anti-Infectives III	Dr. Russell Randall
Oct. 22	20	Anti-Infectives IV (10:30 am- 12:30 pm)	Dr. Russell Randall
Oct. 24	21	Anti-Infectives V (10:30 am- 12:30 pm)	Dr. Russell Randall
Oct. 25	22	Anti-Infectives Clin. Correlation (10:30 am- 12:30 pm)	Dr. Elks
Oct. 29	Tuesday	EXAM III 10:30 - 12:30	
CARDIOVASCULAR/RENAL			
Nov. 06	23	Antihypertensive Drugs	Dr. Eatman-Daniels
Nov. 07	24	Anti-lipidemics (10:30 am - 12:30 pm)	Dr. DeBruyne
Nov. 11	25	Treatment of Ischemic Heart Disease (Anti-anginal Agents)	Dr. Eatman-Daniels
Nov. 14	26	Congestive Heart Failure Drugs	Dr. Eatman-Daniels

Nov. 19	27	Antiarrhythmic Drugs	Dr. Eatman-Daniels
Dec. 02	28	Diuretic Drugs (10:30 am - 12:30 pm)	Dr. Eatman-Daniels
Dec. 17	Tuesday	EXAM IV (Comprehensive) 10:30 a.m. - 12:30 p.m.	

Spring Semester			
RESPIRATORY/PSYCH/HEME			
Jan. 07	29	Respiratory Drugs	Dr. Eatman-Daniels
Jan. 9	30	Toxicology I	Dr. Shepherd
Jan. 13	31	Toxicology II	Dr. Shepherd
Jan. 16	32	Antipsychotic Drugs	Dr. Shepherd
Jan. 23	33	Antidepressants	Dr. Shepherd
Jan. 27	34	Drugs for Anxiety & Insomnia	Dr. Shepherd
Jan. 28	35	Anticoagulants	Dr. Russell Randall
Feb. 03	Monday	EXAM V (10:30 a.m. - 12:30 p.m.)	
GI/ENDOCRINE			
Feb. 05	36	Gastrointestinal Drugs (10:30 am - 12:30 pm)	Dr. Cropp
Feb. 13	37	Drugs of Abuse I	Dr. Russell Randall
Feb. 18	38	Drugs of Abuse II (10:30 am - 12:30 pm)	Dr. Russell Randall
Feb. 25	39	Endocrine Pharmacology I	Dr. Elks
Feb. 27	40	Endocrine Pharmacology II	Dr. Elks
Mar. 04	41	Endocrine Pharmacology III	Dr. Elks
Mar. 06	42	Endocrine Pharmacology IV	Dr. Elks
Mar. 12	Wednesday	EXAM VI 10:30 a.m. - 12:30 p.m.	
MUSCULOSKELETAL / NEURO			
Mar. 27	43	Drugs Used in Treatment of Gout/Arthritis (DMARDS)	Dr. Eatman-Daniels
Mar. 28	44	Local Anesthetics	Dr. Cropp/Dr. Shepherd
Mar. 31	45	Inhaled Anesthetics	Dr. Cropp/Dr. Shepherd
Apr. 02	46	Intravenous Anesthetics	Dr. Cropp/Dr. Shepherd
Apr. 03	47	Opioid Analgesics	Dr. Randall
Apr. 08	48	Drugs Used to Treat Migraines	Dr. Eatman-Daniels
Apr. 10	49	Drugs for Parkinson's / Neurodegenerative Diseases	Dr. Shepherd
Apr. 11	50	Anti-epileptic Drugs	Dr. Sloviter
Apr. 15	51	Drugs Used to Treat Ocular Disorders	Dr. Russell Randall
Apr. 17	52	Chrono-Pharmacology	Dr. Tosini
Apr. 24	Thursday	EXAM VII (Comprehensive) 10:30 a.m. - 12:30 p.m.	
May 14	Wednesday	PHARMACOLOGY MINI-BOARD 9:00 a.m.	

*- Exam times may change, so please pay attention to the MD2 master schedule.